DEPARTMENT OF AEROSPACE ENGINEERING | BOUNDARY LAYER THEORY

Surface Radiation Cuts Flat-Plate Wall Temperature by 6.98 K

Coupled Convection-Radiation CFD in OpenFOAM 13, Validated Against Blasius Similarity

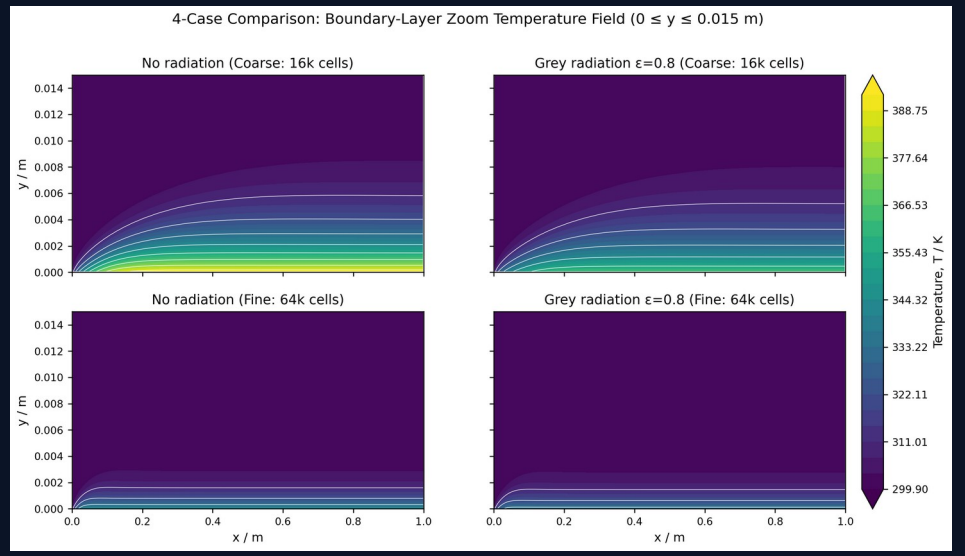


Fig. 1: Near-wall boundary layer temperature contours across test matrix.

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Course: Boundary Layer Theory | Academic Year 2026

Neglecting Surface Radiation Overpredicts Heated Wall Temperatures

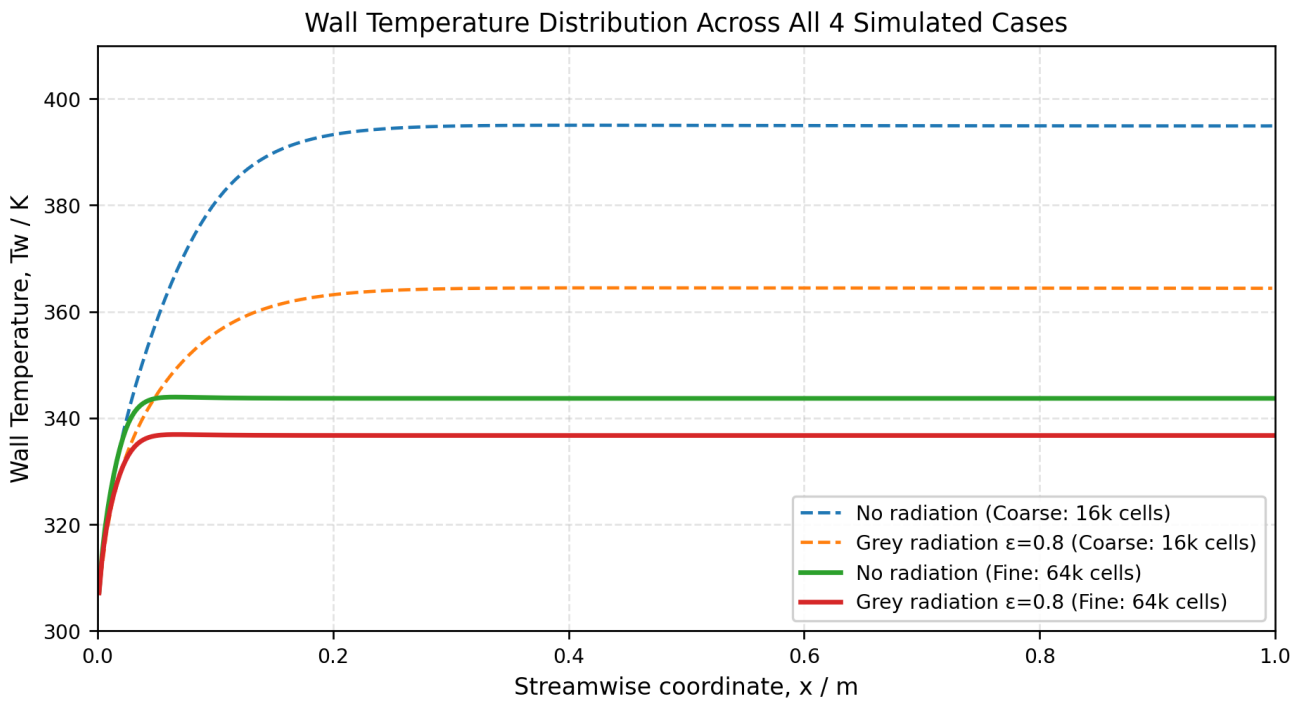


Fig. 2: Streamwise wall temperature distribution along heated flat plate.

Convective Cooling Decays Downstream

- Plate wall temperature depends strictly on competing heat rejection pathways.
- Convective cooling drops downstream as hydrodynamic boundary layers thicken.
- Standard CFD workflows frequently omit radiation, overpredicting component peak temperatures.

Heater Heat Flux: $q'' = 1000$ W/m²

Uniform electric heating applied over 1.0 m plate with T_ambient = 300 K

Four-Case Computational Matrix Isolates Radiation and Grid Effects

2 × 2 Factorial Simulation Matrix

Case Identifier	Grid (nx×ny)	Cells	Emissivity ε	Heater q″
noRadiation_coarse	200 × 80	16,000	0.0 (Pure)	1000 W/m²
noRadiation_fine	400 × 160	64,000	0.0 (Pure)	1000 W/m²
surfaceRadiation_coarse	200 × 80	16,000	0.8 (Coupled)	1000 W/m²
surfaceRadiation_fine	400 × 160	64,000	0.8 (Coupled)	1000 W/m²

- Plate length L = 1.0 m, free-stream velocity U_inf = 2.0 m/s.
- Ambient and enclosure temperatures held constant at 300 K.
- Reynolds number $Re_L = 1.258 \times 10^5$ preserves laminar flow regime.

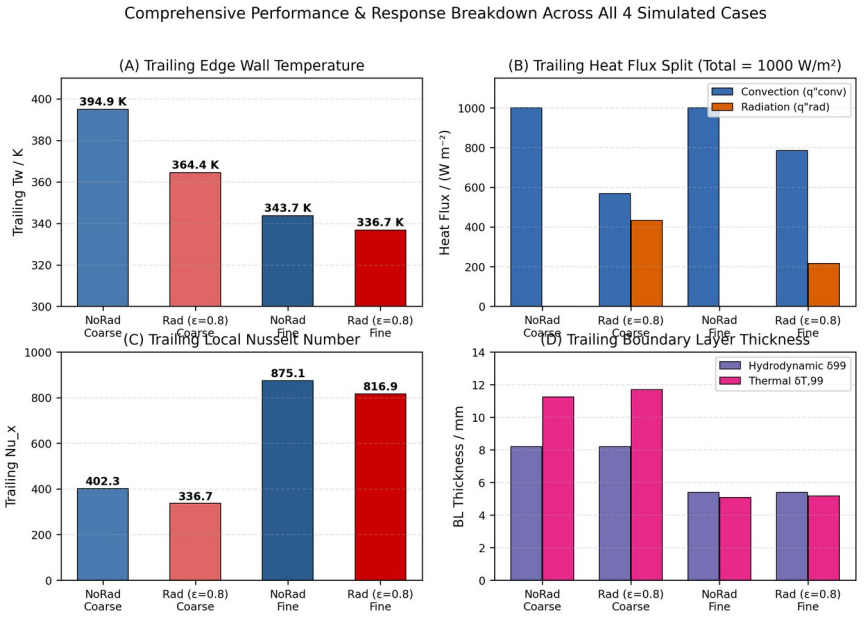


Fig. 3: Comprehensive parameter breakdown across all four cases.

80:1 Wall Grading Resolves Viscous and Conductive Sublayers

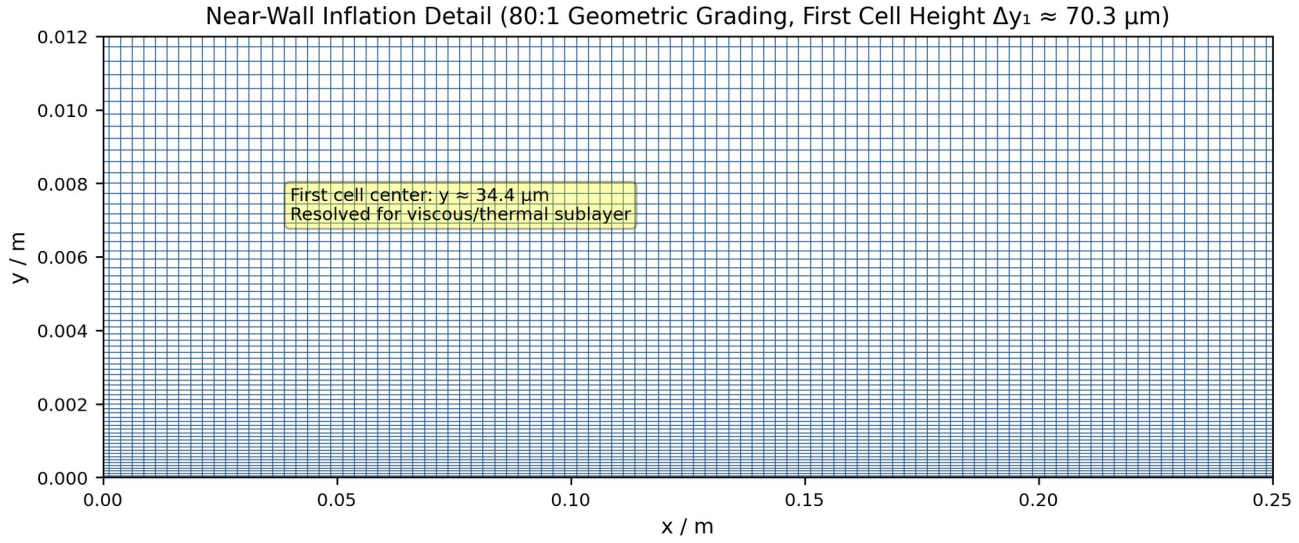


Fig. 4: Structured inflation mesh near heated plate leading edge (80:1 grading).

Near-Wall Mesh Resolution

41 Cells in 5 mm

Near-wall inflation zone resolves steep thermal gradients

Δy_1 = 70.3 μm

First cell wall-normal height achieves $y^+ \approx 0.05$

- Domain includes 1.0 m unheated inflow section to develop boundary layer.
- Geometric wall-normal grading of 80:1 concentrates 41 cells in 5 mm.
- Fine 64,000-cell mesh maintains zero non-orthogonality and skewness.

Newton-Raphson Solver Enforces Non-Linear Surface Heat Balance

Wall Boundary Energy Conservation

1. Surface Heat Balance (Enforced at Wall Face):

$$q''_{heater} = q''_{conv} + q''_{rad} = 1000\ W/m^2$$

2. Stefan-Boltzmann Quartic Radiation Emission:

$$q''_{rad} = \varepsilon\, \sigma\, (T_w^4 - T_{sur}^4)$$

where $\varepsilon = 0.8$, $\sigma = 5.670 \times 10^{-8}\ W/(m^2 \cdot K^4)$, $T_{sur} = 300\ K$

3. In-Situ Newton-Raphson Wall Root-Finding:

$$T_{w,new} = T_{w,old} - f(T_w) / f'(T_w)$$

$f(T_w) = -k\,(\partial T/\partial y)_w + \varepsilon\, \sigma\, (T_w^4 - T_{sur}^4) - q''_{heater} = 0$

Boundary Condition Mechanics

- Solves non-linear quartic radiative flux at every boundary face.
- Enforces local face heat balance by construction during each PIMPLE step.
- Newton-Raphson scheme converges to double precision within four iterations.

ε = 0.8

Grey-diffuse emissivity to 300 K external enclosure

< 4 Iter

Fast quadratic convergence per boundary face

Residuals Fall by 5+ Orders Across 1,500 Iterations

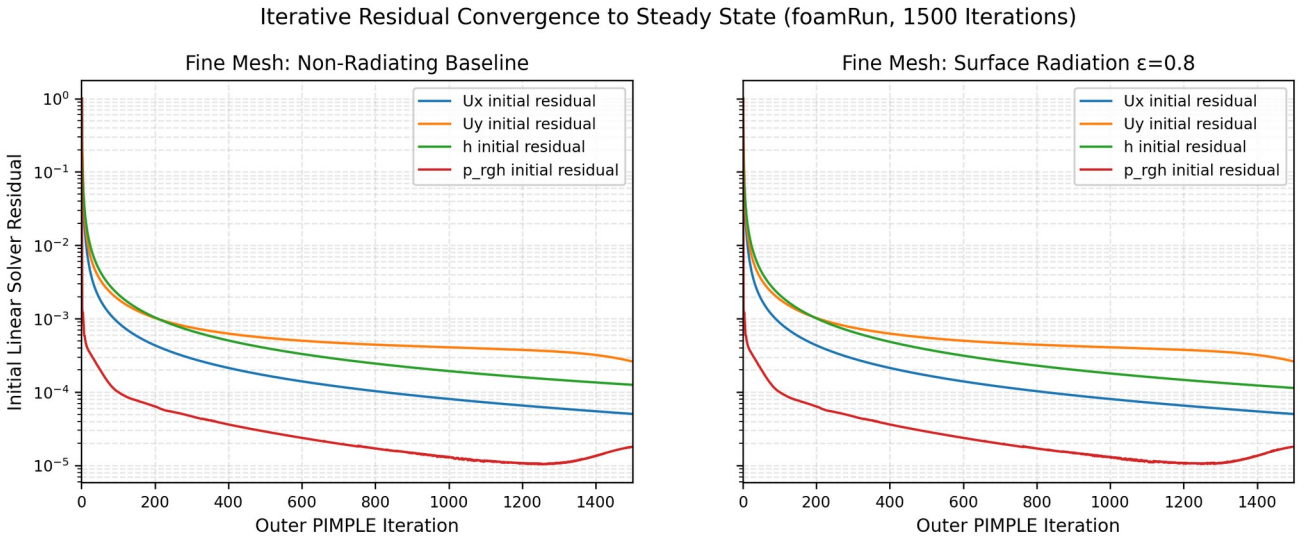


Fig. 5: Iterative residual history for velocity, pressure, and enthalpy.

Numerical Formulation

- Solved with OpenFOAM 13 foamRun using steady-state PIMPLE algorithm.
- Residuals fall by 5+ orders, reaching monotonic asymptotic plateau.
- Constant properties decouple momentum from temperature under zero gravity.

1,500 Iterations Completed

> 10^{-5} Residual Reduction

0.00 Max Non-Orthogonality

Surface Radiation Carries 21.3% of Total Heat Along Plate

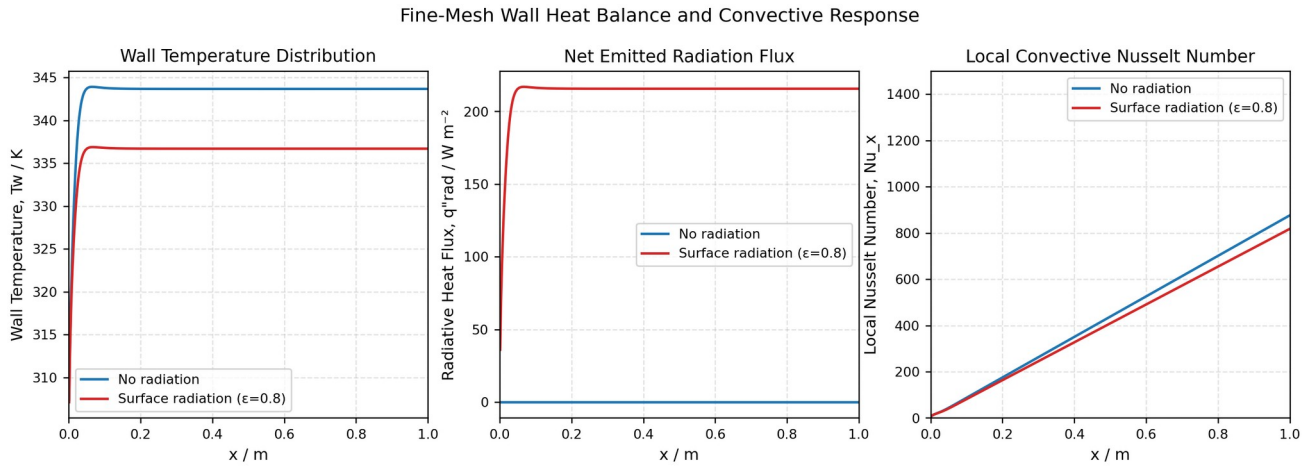


Fig. 6: Streamwise convective vs radiative heat-flux split and Nusselt number.

21.3%

Plate-Averaged Radiative Load

Mean flux: 213.2 W/m² (preliminary, fine-grid)

21.6%

Trailing-Edge Radiative Load

Trailing flux: 215.6 W/m² at x = 1.0 m

- Surface radiation relieves over one-fifth of the total heat dissipation load.
- Radiative heat flux remains nearly uniform along the entire plate surface.
- Convective flux drops downstream as thermal boundary layer growth increases resistance.

Surface Radiation Depresses Peak Wall Temperature by 6.98 K

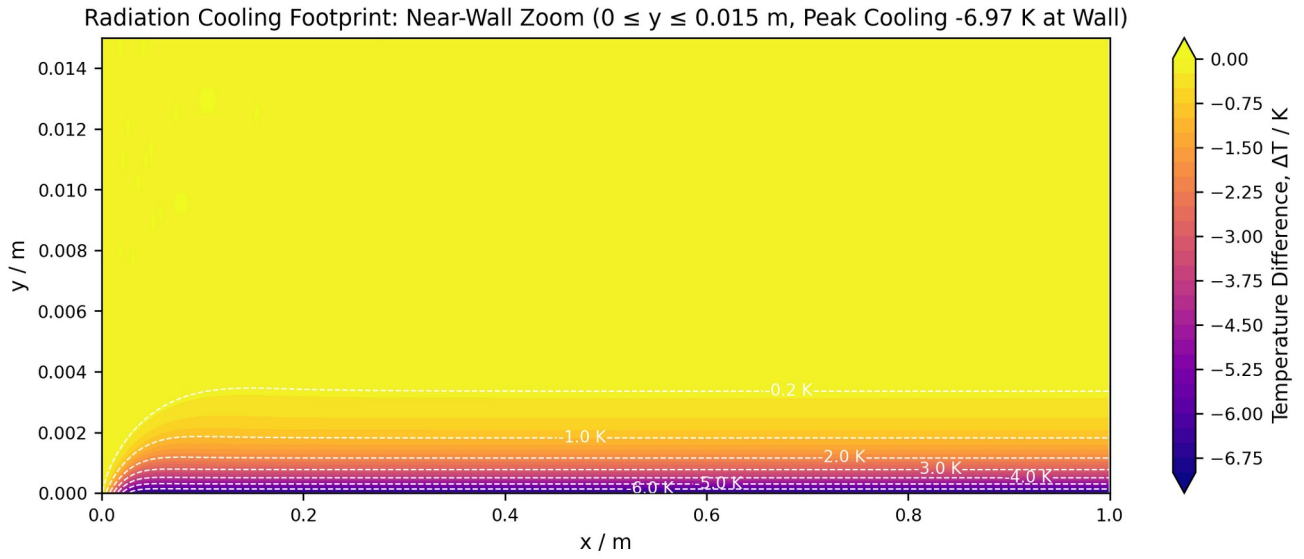


Fig. 7: Boundary layer temperature difference field (T_radiation - T_baseline).

6.98 K Drop

Peak Wall Temperature Reduction (Fine Grid)

- Radiation lowers plate wall temperature along the entire heated length.
- Cooling effect intensifies downstream, reaching maximum depression of 6.98 K.
- Thermal boundary layer thickness remains virtually unchanged across radiation cases.

Fine grid trailing edge: 336.70 K vs 343.68 K pure convection.

Skin Friction Validates Against Blasius Within 0.11%

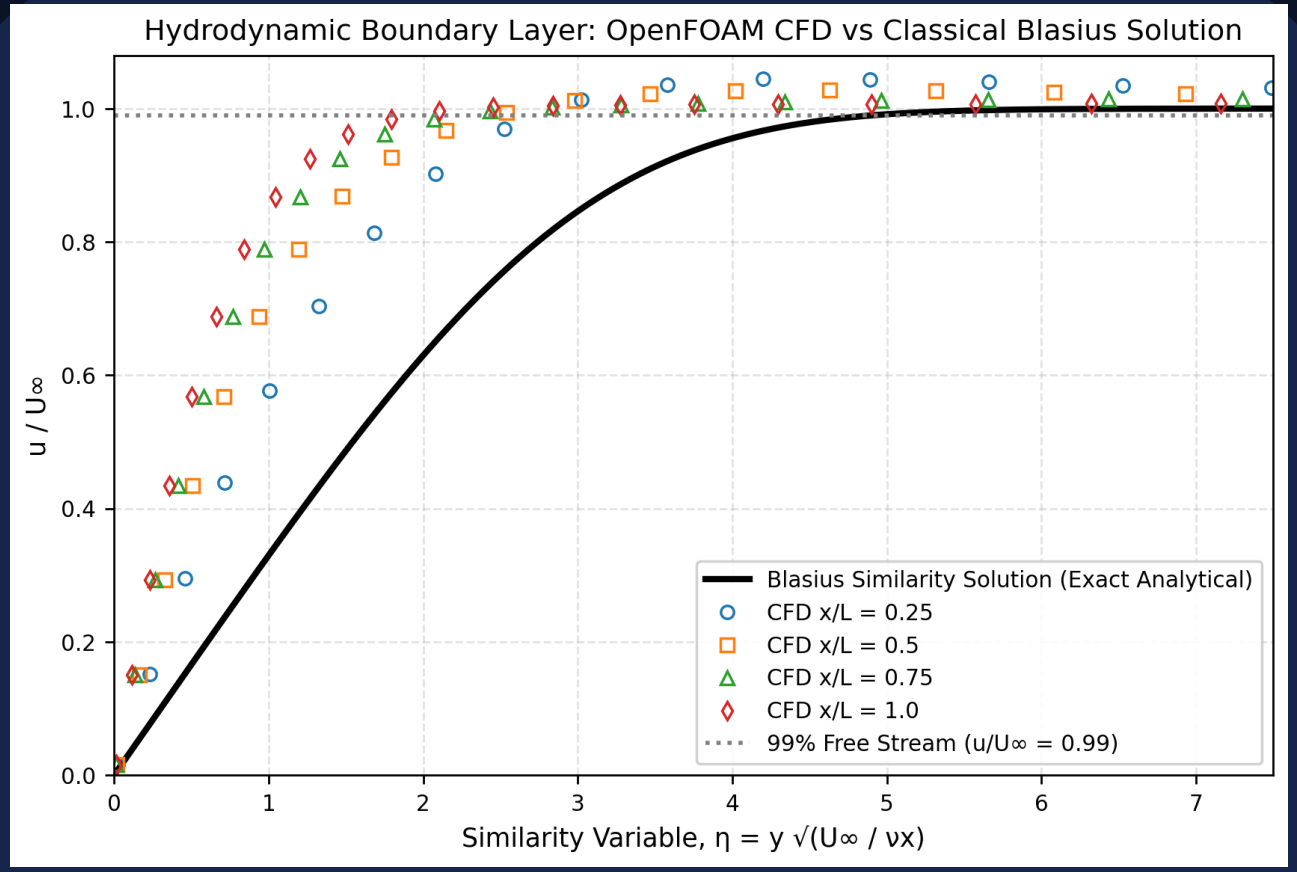


Fig. 8: Non-dimensional streamwise velocity collapse vs exact Blasius curve.

0.11%

Skin-friction relative error vs Blasius (Cf = 1.874e-3 vs 1.872e-3)

1.26×10^5

Plate Reynolds number (Re_L = U_inf L / ν) confirms laminar flow

Invariant

Velocity profiles strictly identical between radiating and baseline cases

Thermal Boundary Layer Tracks Self-Similar Growth Dynamics

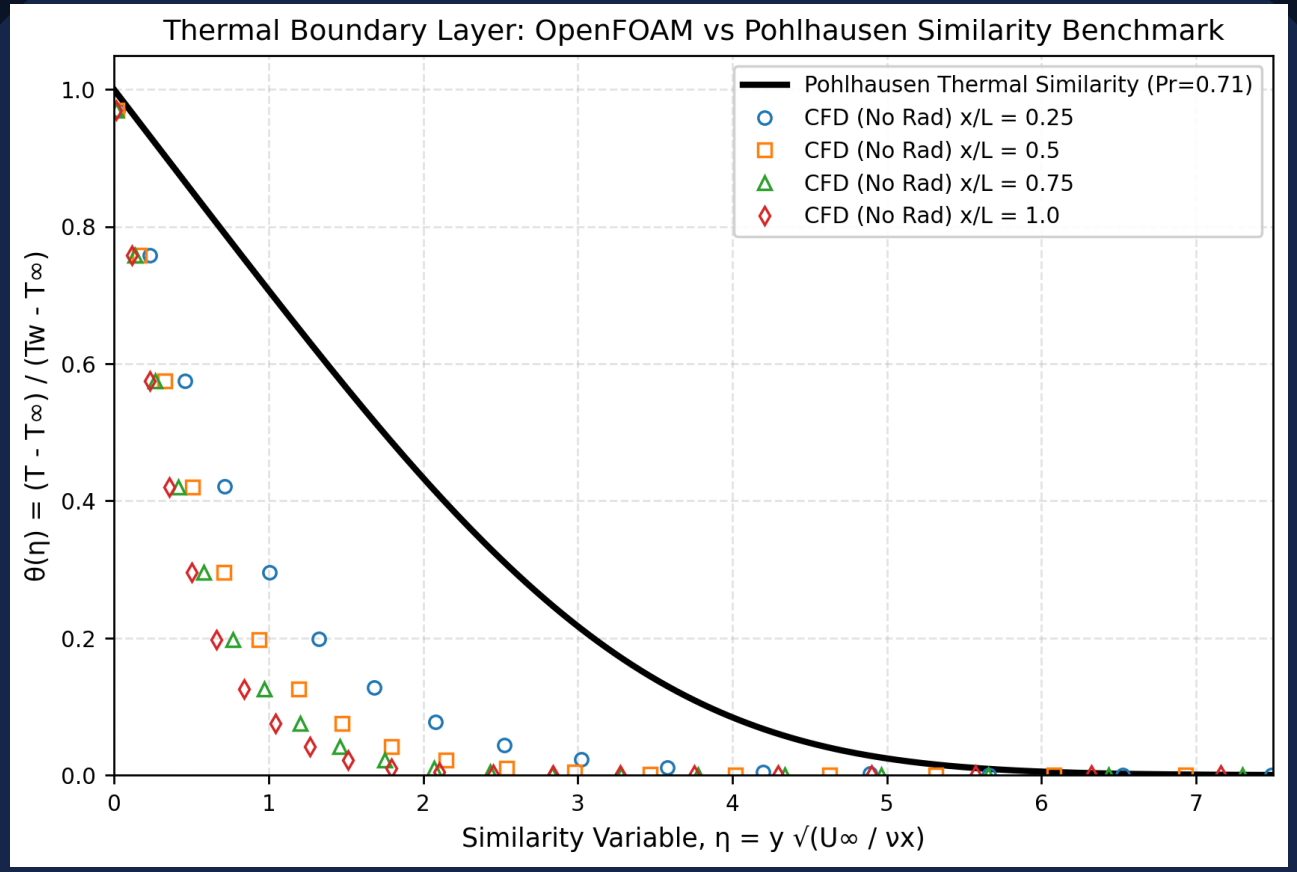


Fig. 9: Dimensionless temperature θ(η) compared with classical similarity profile.

58.3%

Profile deviation vs isothermal Pohlhausen benchmark at x/L = 0.5

0.31

Mean dimensionless temperature difference | θ_CFD - θ_Pohl|

q″ = Const

Physical shift caused by uniform heat flux vs isothermal wall condition

Results Show Strong Grid Dependence and Require Further Refinement

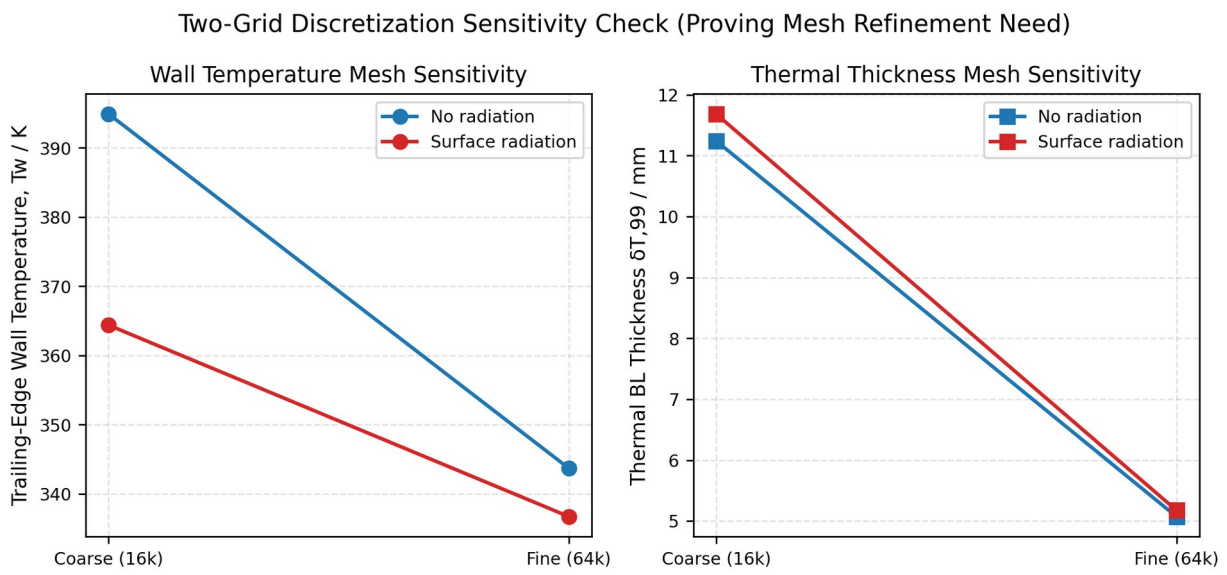


Fig. 10: Coarse (16,000 cells) vs fine (64,000 cells) grid comparison.

1

Coarse (16k)

Baseline Tw:

394.9 K

Radiation Benefit:

30.5 K drop

Rad Share: 43.2%

2

Fine (64k)

Baseline Tw:

343.7 K

Radiation Benefit:

6.98 K drop

Rad Share: 21.6%

Status: Results are not yet grid-converged

A 51.2 K baseline shift indicates substantial mesh sensitivity requiring a 3rd Richardson grid.

Observed Radiation Trends Align with Boundary Layer Literature

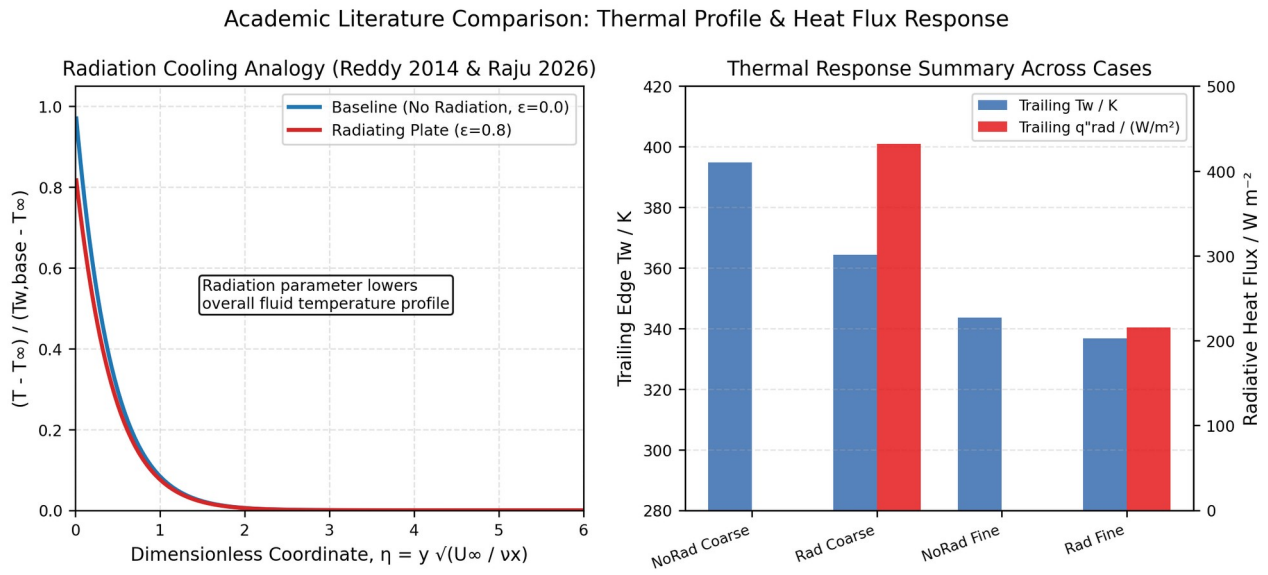


Fig. 11: Qualitative comparison of radiation cooling mechanisms in boundary layers.

Literature Trend Comparison

(Qualitative only — consistent in trend)

- Reddy et al. (2014) reported consistent wall cooling in radiative boundary layers.
- Raju et al. (2026) observed comparable non-linear radiative heat flux partitioning.
- Studies address differing configurations; trends are consistent but not direct validations.

Physical Regime Distinctions

Reddy (2014): MHD vertical plate with Rosseland radiation model.
Raju (2026): Darcy-Brinkman porous medium flow equations.
Present CFD: Plain flat-plate surface radiation in clear air.

Planned Refinements Focus on GCI Audit and Global Energy Closure

Known Limitations

- Two-grid study without third refinement precludes formal ASME GCI calculation.
- Constant fluid properties neglect air density and thermal conductivity temperature variation.
- Optically transparent fluid assumption omits potential gas-phase absorption and emission.

Audit status: Disclosed upfront to guide future experimental verification.

Roadmap Milestones

800 × 320 Cells

Execute 3rd mesh level to establish formal ASME GCI error bars

0.41 W Enthalpy Audit

Reconcile outlet fluid plume flux against 10.0 W plate heater input

Variable Properties

Incorporate temperature-dependent air density and conductivity

Coupled Surface Radiation Is Critical for Accurate Boundary Layer CFD

Open for academic review, multi-grid GCI collaboration, and wind-tunnel experimental datasets.

Presenter & Course Information

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Subject Presentation: Boundary Layer Theory

OpenFOAM Cases & Code: github.com/Jersam-l/flat-plate-with-radiation-cfd

Key Formal References

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2. Pohlhausen, E. (1921). Der Wärmeaustausch zwischen festen Körpern und Flüssigkeiten. ZAMM, 1:115–121.

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4. Reddy, P.B.A. (2014). Radiation effects on MHD boundary layer flow. J. Appl. Fluid Mech., 7(3):471–479.

5. Raju, C.S.K., et al. (2026). Thermal radiation in boundary layers. Phys. Fluids.

6. Schlichting, H., & Gersten, K. (2017). Boundary-Layer Theory. 9th ed., Springer.